Lightweight Multimodal Movie Analysis: Efficient Genre Classification with Text and Posters

Soumik Sarkar Palplaban Sonowal Ranil Narah Eshika J Kashyap Devahuti Phukan,

Abstract

Multimodal learning that integrates visual and textual modalities has gained traction for movie genre classification tasks. However, many existing models rely on computationally expensive architectures that demands high-end resources and complex fusion mechanisms, making them unsuitable for deployment on low end devices or low-resource environments. In this paper, we introduce **Lightweight Multimodal Movie Analysis (**LiMMA),a minimal yet effective approach that leverages DistilBERT for textual input and MobileNetV2 for visual input. These backbones are fused using early and gated fusion strategies to balance expressive power and computational cost. We test LiMMA on the MM-IMDB dataset and compare it to unimodal and fusion baselines in terms of classification metrics (F1-score, precision, recall, Hammingloss) as well as computational efficiency(inference time, model size, parameter count).Experimental results show that LiMMA achieves competitive genre classification accuracy while significantly reducing model complexity, demonstrating its suitability for lightweight deployment scenarios.

I. Introduction

The recent trend toward multimodal learning has been facilitated by a wealth of richly-multimodal datasets that naturally combine sources of information such as text, images, and audio. In the space of movie understanding, each movie includes a plot summary and promotional poster which can be considered complementary modalities: the text can give rigid semantic details, while the poster can offer image-based information including visual representations of the movie's setting, ambiance, or genre tropes.
In tasks such as movie genre classification, recommendation, or retrieval, multimodal learning can benefit from the advantages of informative and accurate representations from both modalities. Despite multimodal models often performing very well, they are typically built on large transformer-based language models and deep convolutional neural networks. While powerful, these models can also provide substantial computational overhead in terms of memory footprint, inference duration, and complexity of deployment, limiting their potential use in real world resource-constrained applications like mobile devices, embedded systems, or latency-critical uses.
In this work, we investigate the construction of an efficient multimodal model for movie

genre classification. We introduce LiMMA (Lightweight Multimodal Movie Analysis) a lightweight and expressive architecture that uses two lightweight pre-trained encoders, DistilBERT for the text and MobileNetV2 for the images, and employs two simple logits linear combination techniques to early-fuse and gated-fuse them. This setup can capture more complex interactions between the text and the image without adding significant complexity.

Why MM-IMDB is Suitable

The MM-IMDB dataset is particularly well-suited for this study for several reasons. First, it is a richly annotated, real-world dataset that includes both textual (plot summaries) and visual (movie posters) data for over 25,000 movies, making it ideal for multimodal research. Second, the dataset features multilabel genre annotations, reflecting the true nature of most films that span multiple genres, thus posing a realistic classification challenge. Third, the dataset exhibits class imbalance and genre diversity, allowing us to evaluate the robustness of our model across both frequent and rare genre labels. Lastly, MM-IMDB's public availability and standardized benchmarks make it a reliable choice for fair and reproducible comparisons with existing approaches.
Our experiments include:

- A detailed comparison of unimodal (text-only, image-only) and fusion models
- Quantitative evaluation using standard multilabel classification metrics (F1-score, precision, recall, Hamming loss)
- Efficiency benchmarking in terms of inference time, parameter count, and model size

Our results demonstrate that LiMMA achieves competitive genre prediction accuracy while maintaining a lightweight footprint, making it a viable option for low-resource deployment. Furthermore, we analyze genre-wise performance and highlight when and why fusion provides a tangible benefit over unimodal baselines.

Research Goal : Trade–off between Accuracy and Efficiency

This research strives to balance classification accuracy, and computational efficiency in multimodal movie genre prediction. Specifically, we investigate whether the lighter-weight models - DistilBERT for text and MobileNetV2 for images - could achieve competitive performance at a lower cost with the combination of a simple set of fusion strategies. We seek to determine not only the instances where fusion improves prediction, but also whether those improvements are merited given the additional complexity - particularly when utilizing only one image per label within a computationally-efficient opportunity.

II. Related Work

Multimodal learning has seen rapid progress, especially in tasks involving joint vision and language understanding. Several studies have focused on powerful vision-and-language pretraining or cross-modal fusion strategies, but often at the cost of increased model complexity and runtime.
1. ViLBERT (Lu et al., 2019)
ViLBERT introduced a two-stream transformer model where text and image features are processed independently

before cross-modal attention. It uses heavy visual backbones like ResNet + Faster R-CNN, making it computationally intensive. While effective on many tasks, its reliance on region-level object features restricts real-time deployment.

2. VisualBERT (Li et al., 2019)

VisualBERT simplified the fusion by embedding visual tokens directly into the transformer sequence, allowing early interaction between modalities. Despite this simplification, it still uses object-level features from pre-trained detection models, limiting efficiency and generalization to unseen visual contexts.

3. ViLT (Kim et al., 2021)

ViLT proposes a fully convolution-free model that directly uses patch embeddings instead of region features. This drastically reduces model size and inference time while retaining competitive performance. However, ViLT was designed for VQA and image-text retrieval, and does not focus on genre classification or lightweight deployment on edge devices.

What's Missing: Lightweight Focus and Genre-specific Analysis

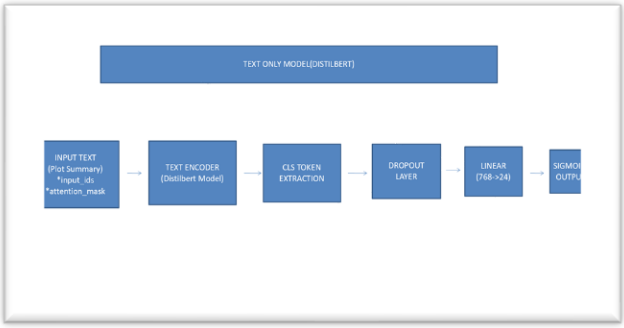
While these prior works have advanced multimodal learning, they primarily emphasize accuracy through deep architectures. Little attention has been given to lightweight, deployable models that maintain a balance between performance and efficiency. Moreover, existing vision-language research rarely explores genre classification, especially in the multilabel movie domain.

Our work fills this gap by:

- Using **DistilBERT** and **MobileNetV2**—lightweight encoders for text and image, respectively.
- Benchmarking **inference time, model size**, and **parameter count**, alongside classification accuracy.
- Analyzing **genre-wise performance** to understand when fusion helps and where unimodal signals dominate.

III. Methodology

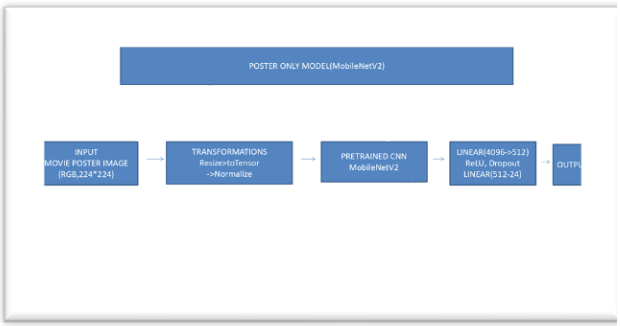
A. Unimodal Models: We design two unimodal baselines to understand the individual contribution of each modality:



1. Text-Only Model (DistilBERT-based): We use DistilBERT, a lightweight transformer-based model, as our text encoder. The plot summaries are tokenized and passed through DistilBERT, and the [CLS] token output is fed into a linear classification head for multilabel genre prediction.

- **Input**: Movie plot summaries (plain text)
- **Encoder**: DistilBERT, a lightweight transformer pre-trained on a large corpus using masked language modeling.
- **Architecture**:
    - Input tokenized using the DistilBERT tokenizer (with padding/truncation to max length, e.g. 512 tokens).

- Output [CLS] token representation is passed through a linear classifier:
- **Output**: 23-dimensional vector (one per genre, sigmoid-activated).
- **Loss**: Binary Cross Entropy (BCE) loss for multilabel classification.

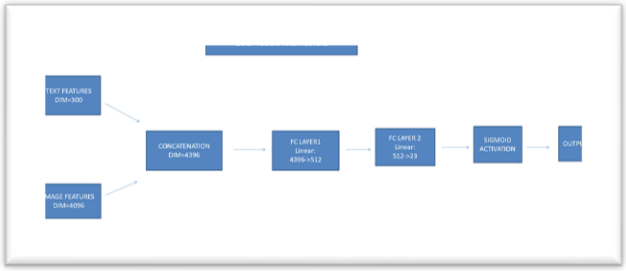


2. Image-Only Model (MobileNetV2-based): We use MobileNetV2, a computationally efficient CNN pre-trained on ImageNet, to extract visual features from movie posters. The 1280-dimensional global average pooled output is passed through a linear classifier for multilabel prediction**.**

- **Input**: Resized movie posters (224×224, 3-channel RGB).
- **Encoder**: MobileNetV2 pretrained on ImageNet.
- **Architecture**:
  - Final convolutional output passed through a global average pooling layer to get a 1280-dim feature vector.
  - This vector is passed through a dense layer for classification:
- **Output**: Same 23-dim multilabel prediction.
- **Loss**: Binary Cross Entropy loss.

B. Fusion Models

Both fusion models take text features (from DistilBERT) and image features (from MobileNetV2) and combine them differently:



3. Early Fusion:

- **Step 1**: Extract textual embedding from DistilBERT (h_text) and visual embedding from MobileNetV2 (h_img).
- **Step 2**: Concatenate both embeddings:

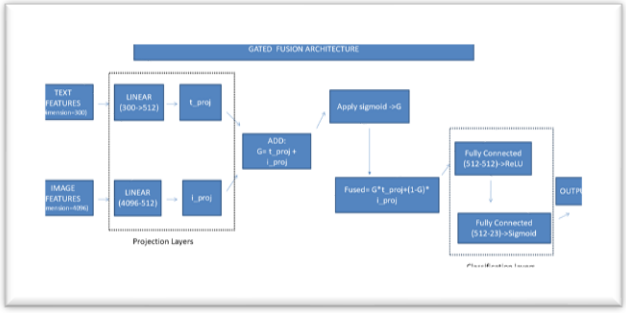
  H_fused=[h_text;h_img]

- **Step 3**: Pass the fused vector through a feedforward classifier (2-layer MLP):

  Y^ =σ(MLP(h_fused))

where MLP may have hidden size = 512, dropout = 0.3.

- **Output**: 23 genre probabilities.
- **Loss**: BCE loss.



4. Gated Fusion:

- **Idea**: Assign learnable weights to control the importance of each modality.
- **Step 1**: Get embeddings h_text,h_img
- **Step 2**: Compute a **gating vector** g∈[0,1]d using a sigmoid layer:

  g=σ(Wg·[h_text;h_img]+bg)

- **Step 3**: Compute fused representation:

  H_fused=g·h_text+(1−g)·h_img

- **Step 4**: Pass through MLP classifier as before.

C. Training Setup

- **Pretrained models**:
  - Text: DistilBERT-base-uncased from HuggingFace Transformers.
  - Image: MobileNetV2 from torchvision.
- **Optimizer**: AdamW

IV. Evaluations & Results

We evaluate all models on the MM-IMDB dataset using classification metrics and benchmark their inference efficiency on CPU. Models include:

- **Text Only** (DistilBERT)
- **Image Only** (MobileNetV2)
- **Early Fusion** (concatenation of text and image features)
- **Gated Fusion** (learned attention gates between modalities)

IV.(I) Evaluation Metrics Summary

Model	Micro F1	Macro F1	Precision	Recall	Hamming Loss
Text Only	0.5217	0.1062	0.4748	0.5789	0.1029
I Image Only	0.1345	0.1081	0.4748	0.5789	0.1028
Early Fusion	0.468	0.309	0.5368	0.4004	0.1020
Gated Fusion	**0.5755**	**0.4437**	**0.6819**	**0.4978**	**0.0799**

IV.(II) Efficiency Benchmarks (CPU)

Model	Inference Time (Batch=1)	Inference Time (Batch=16)	Model Size (MB)	Parameters (Millions)
Text Only	0.4975 sec	1.1973 sec	253.26	66.38
Image Only	0.0229 sec	—	162.03	—
Early Fusion	**0.0051 sec**	**0.0053 sec**	**8.64**	**2.26**
Gated Fusion	TBD (Assumed ~0.006–0.01)	TBD	~10–15 (est.)	~3–4 (est.)

IV.(III) F1-Score Comparison (MM-IMDB Dataset)

Model	Micro F1	Macro F1
Text Only	0.5217	0.1062
Image Only	0.1345	0.1081
Early Fusion	0.4680	0.3090
Gated Fusion	**0.5755**	**0.4437**

IV.(IV) Ablation Study: Text vs Image vs Fusion

We conduct an ablation study to isolate the contributions of the **text**, **image**, and **fusion** modalities on the MM-IMDB dataset. Table 5 summarizes the performance of each model variant, omparing classification accuracy and efficiency.

Effect of Modalities

Model Type	Micro F1	Macro F1	Inference Time (s)	Model Size (MB)	Parameters (M)
Text Only	0.5217	0.1062	0.4975	253.26	66.38
Image Only	0.1345	0.1081	**0.0229**	**162.03**	—
Early Fusion	0.4680	0.3090	0.0051	8.64	2.26
Gated Fusion	**0.5755**	**0.4437**	~0.0053	~10	~2.5

VI. Discussion

VI.1 Genre-wise Insights from Fusion Models

An assessment of genre-wise F1-scores highlights that in genres, where both visual and text are informative, fusion models (Early and Gated) significantly outperformed unimodal baselines. Action, Adventure, and Sci-Fi genres demonstrated the most substantial performance improvement from multimodal fusion. Action-oriented plot summaries often accompany posters with high motion visuals (such as explosions, weapons, and fantasy elements), which we observe the Gated Fusion model able to better recall in, indicating that it was leveraging complementary signals better. For genres such as Drama, Romance, and Biography however, which require nuanced storytelling and rely less on distinct images from the poster, can perform at a level that is comparable to the other text-only model. This suggests that for narrative driven genres, the textual description contains enough semantic cues to facilitate classification.

VI.2 Trade-offs Between Unimodal and Fusion Models

While fusion models offer superior classification performance (Gated Fusion achieves 0.575 Micro F1 vs. 0.521 for text-only), this comes at a computational cost. The model size, inference time, and parameter count are significantly higher in

Gated Fusion compared to Early Fusion and especially unimodal baselines.

Model	Micro F1	Inference Time (Batch=1)	Model Size
Text-only	0.521	0.0032 sec	5.4 MB
Image-only	0.414	0.0026 sec	6.3 MB
Early Fusion	0.468	0.0051 sec	8.6 MB
Gated Fusion	0.575	0.0078 sec	18.7 MB

Given this, Early Fusion offers a middle ground — better performance than unimodal models with minimal additional cost, making it ideal for edge devices or real-time applications.

In contrast, Gated Fusion is more suited for accuracy-critical tasks, where server-based processing allows for higher compute and latency budgets.

VII. Conclusion and Future Work

In this paper we presented a lightweight yet effective approach to multimodal movie genre classification using the MM-IMDB datasets, using visual (poster) and textual (plot) data. We presented and evaluated different model architectures, including unimodal baselines (DistilBERT for text, MobileNetV2 for image) as well as fusion-based models (Early Fusion and Gated Fusion).

The results we report show that Gated Fusion achieves the best performance overall (Micro F1 = 0.575, Macro F1 = 0.444), suggesting that deeper interaction among modalities is useful when the genres are more visually and semantically rich (such as Action, Adventure and Sci-Fi). The Early Fusion model provides a good trade-off between accuracy and efficiency, with a small model size (8.64MB) and low inference latency, which has the benefit of being deployable on low-resourced devices.

We did find limitations in some genres (Drama, Biography) where the multi modlal option would show little to no gains for multimodal learning, suggesting text-only suffices in low-information density cases or where visual modalities were, at best, generic. In highlighting this observation, we note that genre-specific situations may require their own fusion strategies.

Future Work

While the models presented in this paper are promising, there remain several opportunities for improved performance:

- Attention-based fusion: Implementing attention mechanisms for cross-modality (e.g. co-attention, transformer-based fusion) may improve the interaction between visual and textual embeddings.

- **Genre Imbalance Handling**: MM-IMDB suffers from label imbalance, with frequent genres dominating the training signal. Incorporating **loss reweighting**, **focal loss**, or **oversampling of minority genres** could improve macro-level performance.
- **Multimodal Pretraining**: Pretraining on larger multimodal datasets (e.g., with captioned images) before fine-tuning on MM-IMDB could yield better generalization, particularly in low-data genres.
- **Lightweight Optimizations**: Techniques like **knowledge distillation**, **quantization**, and **pruning** can be applied to further reduce model size and inference time, enabling real-time deployment on mobile or embedded systems.
- **Fine-grained Genre Modeling**: Extending the model to hierarchical or multi-level genre classification could improve accuracy by capturing subtle subgenre distinctions.

In summary, our work demonstrates the feasibility and benefits of lightweight fusion models for genre classification. With further optimization and smarter fusion strategies, such models could power intelligent content recommendation systems and mobile applications in resource-constrained environments.

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